Pre-Deployment Checklist

☐ **Batteries**: Ensure the Rover LiPo is fully charged, and the 8BitDo controller is powered on/paired.

☐ **Connection**: Confirm there is a stable WiFi connection shared between your laptop and the Pi.

If WiFi is not available see the next check.

☐ **Environment**: If going deep off-grid, ensure your laptop is broadcasting a Mobile Hotspot and the Pi is configured to automatically connect to it for Phase 4.

☐ **Mission Control**: Plug the Controller Heltec ESP32 into your laptop via USB. Open a terminal on your laptop and run

```
python python\base_station.py
```

> **!! Leave this terminal visible on-screen. This is where live field data will appear.**

Step 1: Mapping the Field (Phase 1)

1. Turn on the rover and SSH into the Raspberry Pi from your laptop.
2. In a separate terminal run the mapping script:

PHASE 1

```
# Bash
~/ros2_ws/demo/01_mapping.sh
```

3. Using the 8BitDo controller, drive the rover manually through the test scene. Drive slowly to allow the Lidar to build a crisp, high-resolution map of the environment.
4. Once the area is mapped, park the rover safely. Run

```
ros2 run nav2_map_server map_saver_cli -f /home/rover/ros2_ws/maps/field_map
```

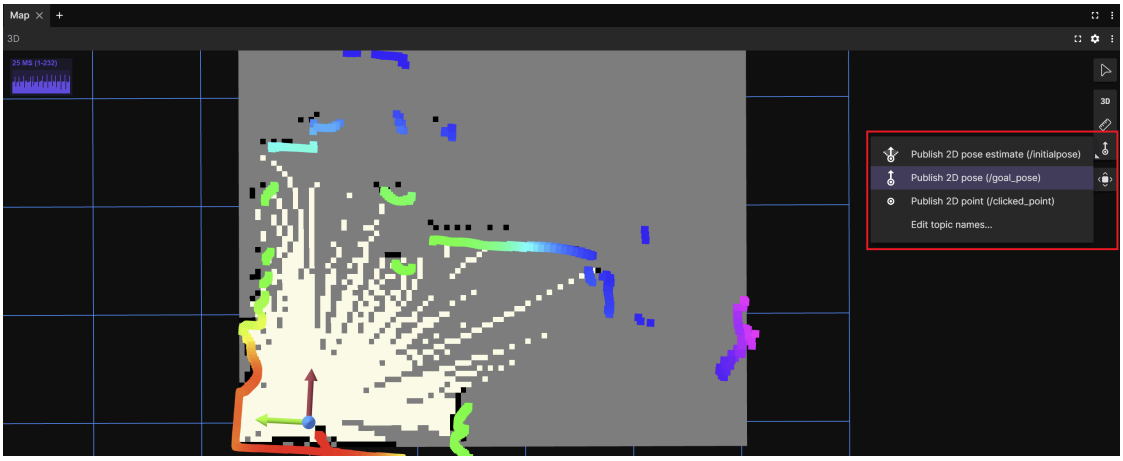
to save the built SLAM map as a pair of files (`.yaml` and `.pgm`).
Do not turn off the rover or kill Phase 1.

Step 2: Foxglove Waypoint Extraction

1. Open Foxglove Studio on your laptop.
2. Open a **Web Socket** connection, select ROS 2, and connect to the Pi using the address: `(ws://<PI_IP_ADDRESS>:8765)`.
3. Add a **3D Panel** to view the live SLAM map.
4. Ensure the **2D Pose** tool is publishing to the `/goal_pose` topic. Run waypoint logger in a separate terminal:

```
python demo/waypoint_logger.py
```

5. Select the **2D Pose** tool from the Foxglove toolbar. Click on the map exactly where you want the rover to park in front of a plant (make sure to point the arrow in the direction you want it to face the plant).



6. Once a point is set, check the terminal for `waypoint_logger.py` to extract the waypoint:

```
[INFO] [1773094267.996522581] [waypoint_logger]: 📍 Waypoint Logger active. Click and drag a
 Nav Goal in Foxglove to generate a coordinate...
failed to create array for field 'prefixes'
--- Copy below this line ---
1.094, 1.177, 0.000
----------------------------
```

7. Repeat this for every plant you want to inspect.

Step 3: Programming the Mission

1. Open a second SSH terminal to the Pi.
2. Open the `route.txt` file.
3. Paste/Type in the (X, Y, θ) coordinates you extracted from the waypoint logger script.
4. Add $(0.0, 0.0, 0.0)$ as the final line so the rover returns to home base. Save and exit.

Step 4: Vision & Autonomy Boot-up (Phases 2 & 3)

1. In a third Pi terminal, start the camera:

PHASE 2

```
# Bash
~/ros2_ws/demo/02_vision.sh
```

3. Open another SSH terminal to the Pi.
4. **CRITICAL**: Press the **'X' button** on the 8BitDo controller. Confirm the OLED screen on your laptop's LoRa module has switched into **AUTO NAV** mode.
5. Launch the autonomy stack in a new terminal:

PHASE 3

```
# Bash
~/ros2_ws/demo/03_autonav.sh
```

Note: This script runs Nav2 to use the map built in phase 1.

Step 5: The Autonomous Patrol

At this point, take your hands off the controller.

1. The rover will drive itself to the first coordinate in `route.txt`.
2. It will stop, pause to let the camera stabilize, and snap a picture.
3. The PlantCV LAB-color algorithm will run, mapping the healthy green vs. diseased tissue.
4. The moment it finishes computing, you will hear/see a blip on your laptop! Look at your `base_station.py` terminal to see the incoming LoRa CSV string populating live.
5. The rover will proceed to the next plant.

Step 6: Data Retrieval (Phase 4)

1. Wait for the rover to finish its patrol and return to `0.0, 0.0` (Home Base), where it is safely back in range of your WiFi or laptop hotspot.
2. On the Pi terminals, press **Ctrl+C** to shut down Phases 1, 2, and 3.
3. Start the local web server in another terminal:

PHASE 4

```bash
# Bash
~/ros2_ws/demo/04_web_server.sh
```

4. On your laptop, open Google Chrome or Firefox and navigate to `http://<PI_IP_ADDRESS>:8000`.
5. Click and download your high-resolution pseudocolored plant maps and your final `field_analysis_log.csv` file!
6. Once downloaded, hit **Ctrl+C** on the web server terminal to shut it down, and power off the rover.